FUNCTIONS

LINEAR FUNCTION-_STRAIGHT LINE GRAPHS – REVISION WORKSHEET

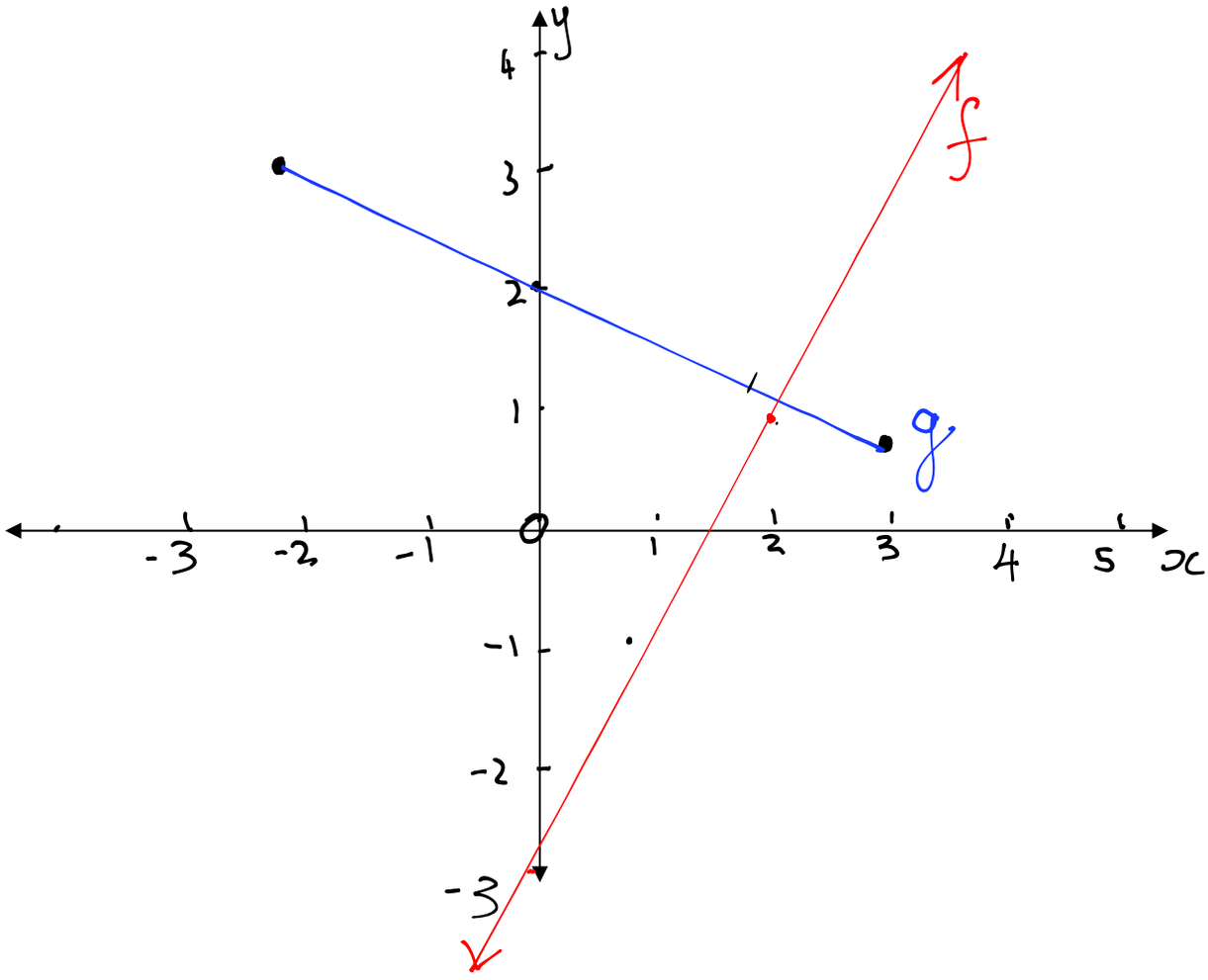
1. Using the **table method** sketch the graphs of the following functions

a) $f: x \to 2x - 3$

x	0	1	2
y	-3	-1	1

b) $g: x \to -\frac{1}{2}x + 2 \quad x \in [-2; 3]$ ← Restricted domain

x	-2	0	3
y	3	2	1/2



2. **i)** Find the x **and** y **intercept** of each of the following
ii) Sketch the graphs using the Dual Intercept method (on the same axes)

a) $2x - y = 3$

x-int (y=0)
2x - 0 = 3
2x = 3
x = 3/2

y-int (x=0)
2(0) - y = 3
y = -3

b) $x - 3y = 5$

x-int:
x - 3(0) = 5
x = 5

y-int
0 - 3y = 5
y = -5/3

c) $4x = 6y + 8$

x-int

$4x = 6(0) + 8$

$4x = 8$

$x = 2$

y-int

$4(0) = 6y + 8$

$-6y = 8$

$y = -\frac{4}{3}$

d) $2y - 3x = 6$

x-int

$2(0) - 3x = 6$

$-3x = 6$

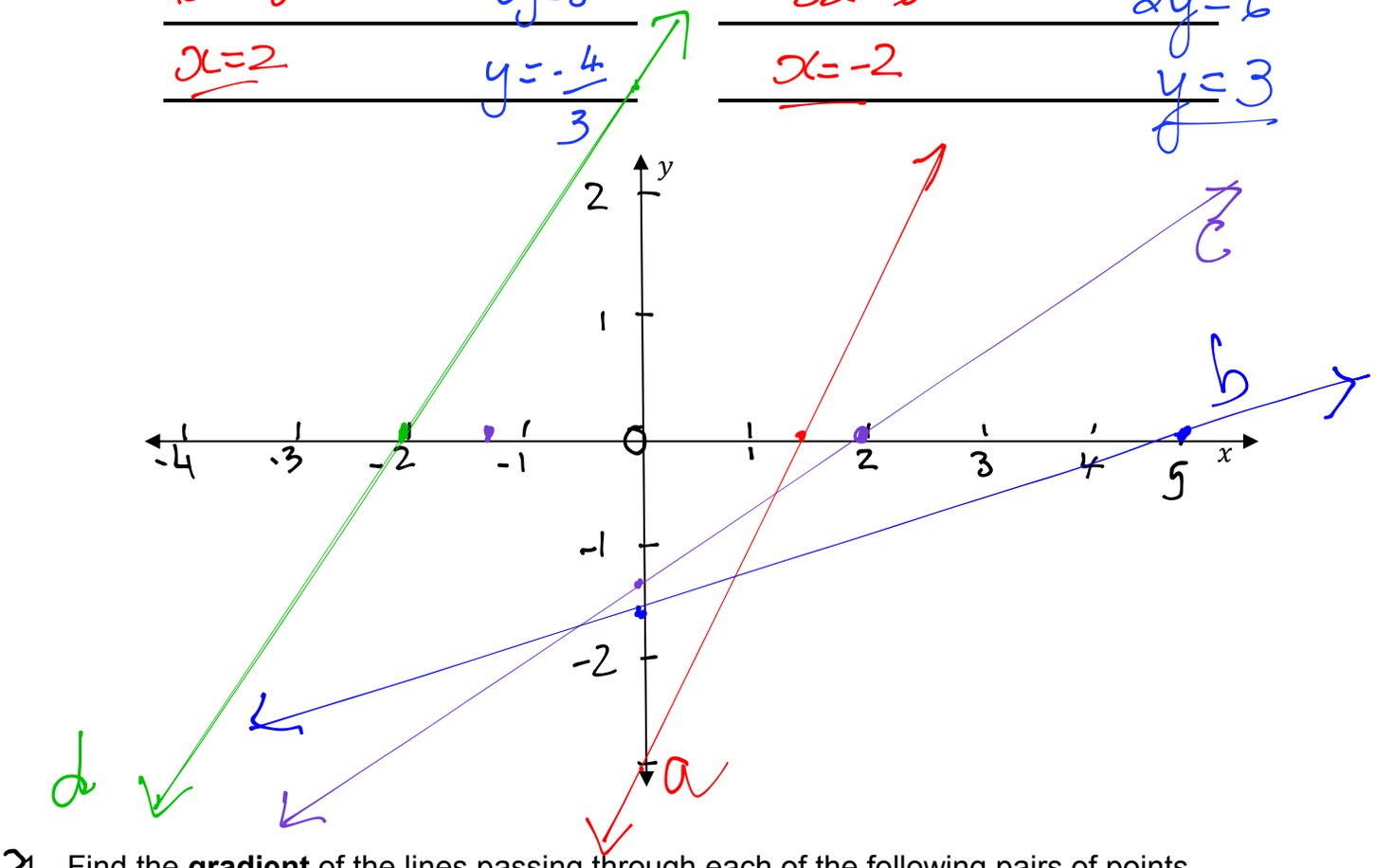
$x = -2$

y-int

$2y - 3(0) = 6$

$2y = 6$

$y = 3$



3. Find the **gradient** of the lines passing through each of the following pairs of points

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

a) $(-2; 5)$; $(1; 11)$

$m = \frac{11 - 5}{1 - (-2)} = \frac{6}{3} = 2$

b) $(2; -5)$; $(-3; 5)$

$m = \frac{5 - (-5)}{-3 - 2} = \frac{10}{-5} = -2$

c) $(-2; -4)$; $(2; 2)$

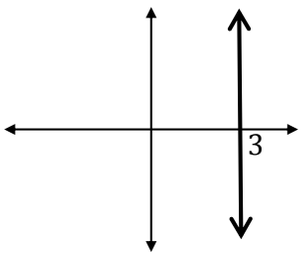
$m = \frac{2 - (-4)}{2 - (-2)} = \frac{6}{4} = \frac{3}{2}$

d) $(-1; 6)$; $(-5; -8)$

$m = \frac{-8 - 6}{-5 - (-1)} = \frac{-14}{-4} = \frac{7}{2}$

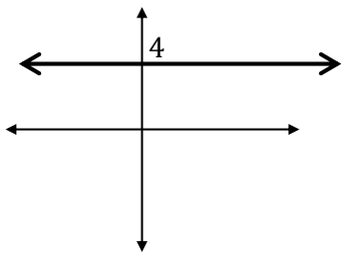
4. Find the equation of the following lines

a.



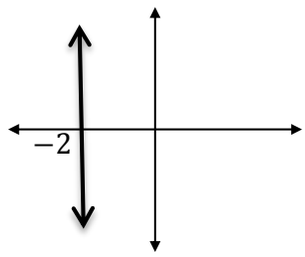
$x = 3$

b.



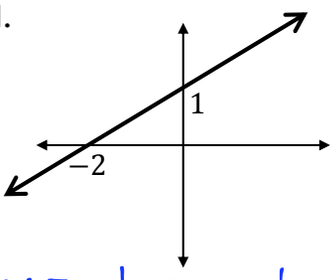
$y = 4$

c.



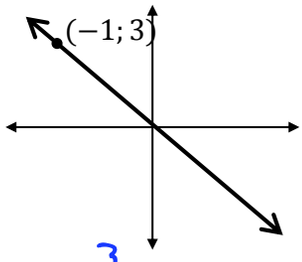
$x = -2$

d.



$y = \frac{1}{2}x + 1$

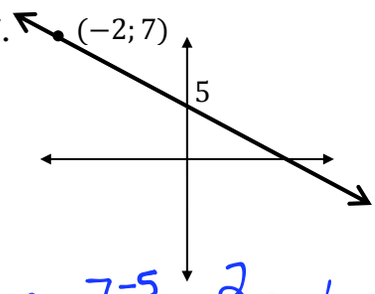
e.



$m = \frac{3}{-1}$

$y = -3x + 0$

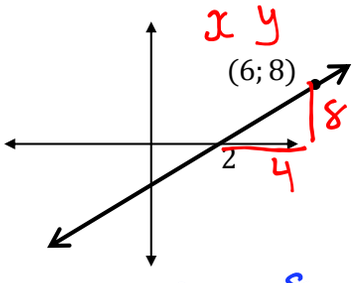
f.



$m = \frac{7-5}{-2-0} = \frac{2}{-2} = -1$

$y = -1x + 5$

g.



$m = \frac{8-0}{6-2} = \frac{8}{4} = 2$

$y = 2x + c$

Sub (6;8)

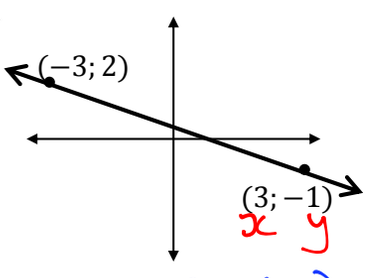
$8 = 2(6) + c$

$8 = 12 + c$

$-4 = c$

$\therefore y = 2x - 4$

h.



$m = \frac{2-(-1)}{-3-3}$

$m = \frac{3}{-6} = -\frac{1}{2}$

$y = -\frac{1}{2}x + c$

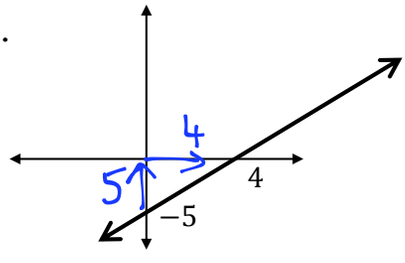
$-1 = -\frac{1}{2}(3) + c$

$-1 = -\frac{3}{2} + c$

$\frac{1}{2} = c$

$y = -\frac{1}{2}x + \frac{1}{2}$

i.



$m = \frac{5}{4}$

$y = \frac{5}{4}x - 5$

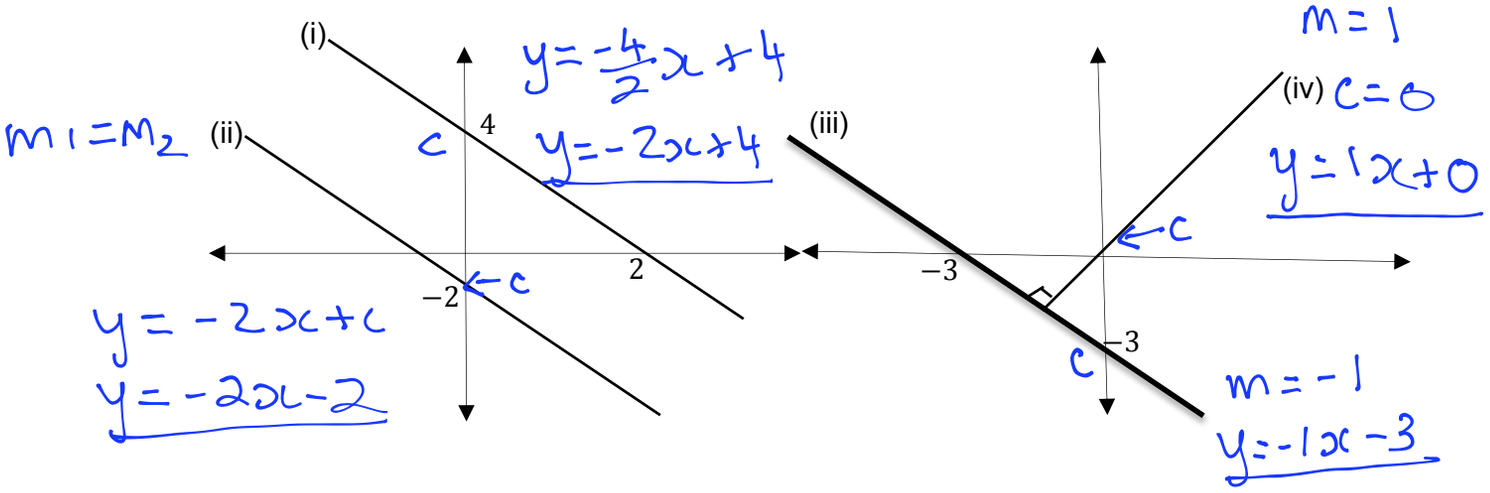
2. Parallel and Perpendicular Lines

a) Parallel lines have Equal gradients $[m_1 = m_2]$

b) Perpendicular Lines: The product of the gradients is equal to -1

$[m_1 \times m_2 = -1]$

c) Determine the equations of graphs (i) – (iv)



SOLUTIONS

x and y intercepts

2a) $x = 1{,}5 \quad y = -3$

b) $x = 5 \quad y = -\frac{5}{3}$

c) $x = 2 \quad y = -\frac{4}{3}$

d) $x = -2 \quad y = 3$

3. (Gradients)

a) 2

b) -2

c) $\frac{3}{2} = 1{,}5$

d) $\frac{7}{2} = 3{,}5$

4. Equations

a) $x = 3$

b) $y = 4$

c) $x = -2$

d) $y = \frac{1}{2}x + 1$

e) $y = -3x$

f) $y = -1x + 5$

g) $y = 2x - 4$

h) $y = -\frac{1}{2}x + \frac{1}{2}$

i) $y = -\frac{5}{4}x - 5$

5. Parallel and Perpendicular lines

i) $y = -2x + 4$

ii) $y = -2x - 2$

iii) $y = -x - 3$

iv) $y = \frac{1}{3}x$

Geogebra Links for the Linear Function:

https://www.geogebra.org/m/sBrHDGF2

Change the "m" and "c" values and see what happens to the graph.

https://www.geogebra.org/m/YHzrmm5R

Investigation of linear graph, with sliders for m and c. Extension to move values of 2nd line such that they are parallel.

Parabola

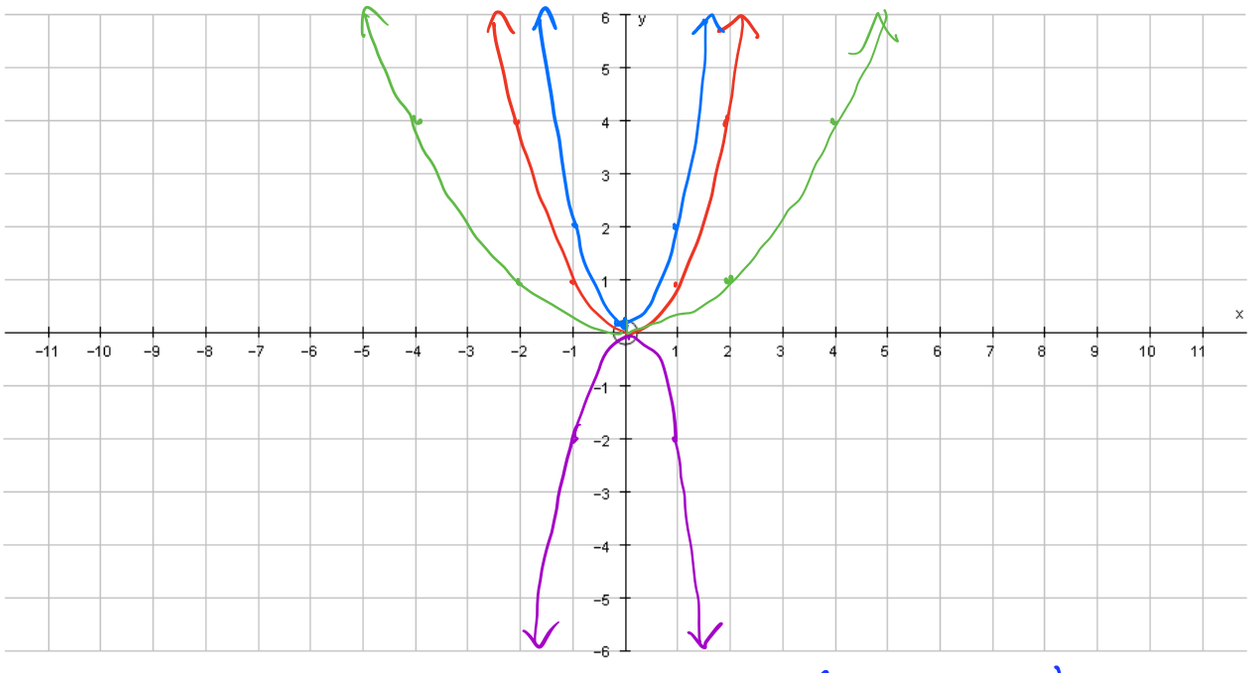
$y = ax^2$ $y = ax^2 + q$

$$y = ax^2$$

A. **Sketch the following graphs on the same set of axes. Use the table to help you. Use different colours to represent each graph**

x	-3	-2	-1	0	1	2	3
$y = x^2$	9	4	1	0	1	4	9
$y = 2x^2$	18	8	2	0	2	8	18
$y = -2x^2$	-18	-8	-2	0	-2	-8	-18
$y = \frac{1}{4}x^2$	9/4	1	1/4	0	$\frac{1}{4}$	1	9/4

i. $y = x^2$ ii $y = 2x^2$ iii. $y = -2x^2$ iv. $y = \frac{1}{4}x^2$

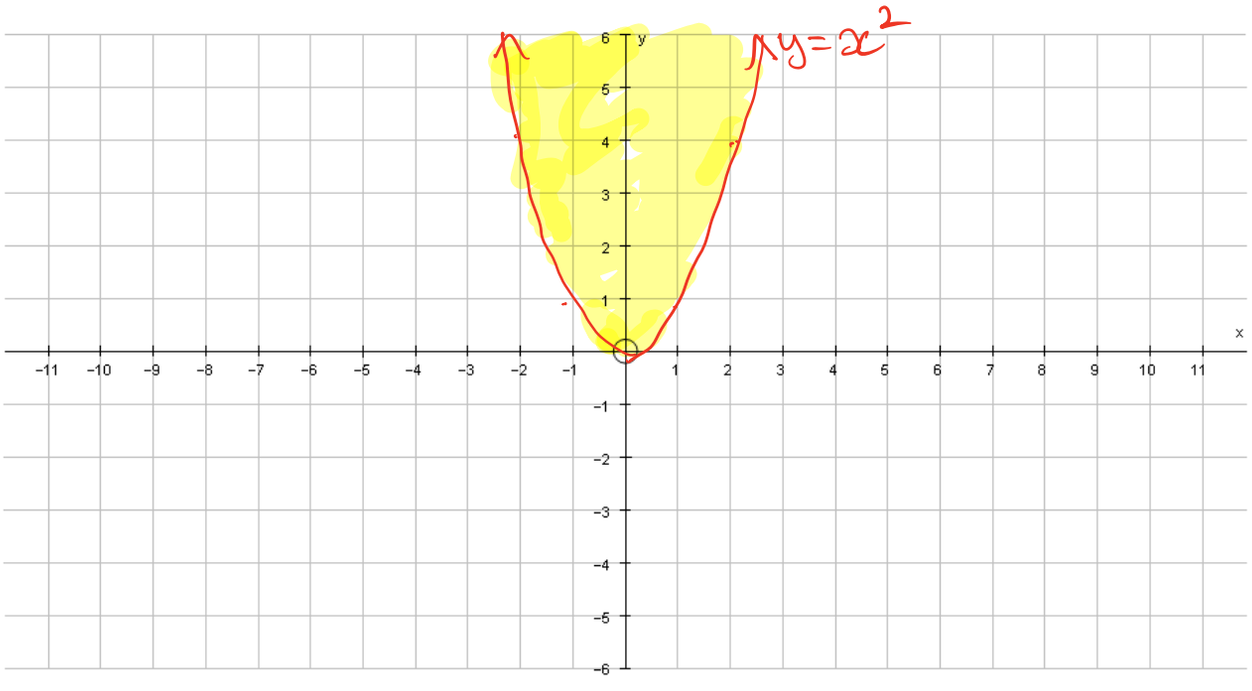


How does the value of ***a*** affect the graph? a > 0 ∴ ∪ min value
a < 0 ∴ ∩ max value . Determines shape wide/narrow

Are the graphs of $y = 2x^2$ and $y = -2x^2$ reflections of each other? yes

What is the equation of the reflection line? y = 0 (the x-axis)

	$y = x^2$	$y = 2x^2$	$y = -2x^2$	$y = \frac{1}{4}x^2$
Value of *a*	1	2	-2	1/4
X – intercept	0	0	0	0
Y – intercept	0	0	0	0
Domain	$x \in R$	$x \in R$	$x \in R$	$x \in R$
Range	$y \in R : y \geq 0$	$y \in R : y \geq 0$	$y \in R : y \leq 0$	$y \in R : y \geq 0$
Co-ordinates of the turning point	(0 ; 0)	(0 ; 0)	(0 ; 0)	(0 ; 0)
Axis of Symmetry	$x = 0$	$x = 0$	$x = 0$	$x = 0$



i. Sketch the graph of $y = x^2$ and indicate on the graph where $y \geq x^2$.

ii. Calculate the average gradient of the curve between x = 2 and x = 4.

$m = \frac{y_2 - y_1}{x_2 - x_1}$ (2 ; 4) and (4 ; 16)

x_1 y_1 x_2 y_2

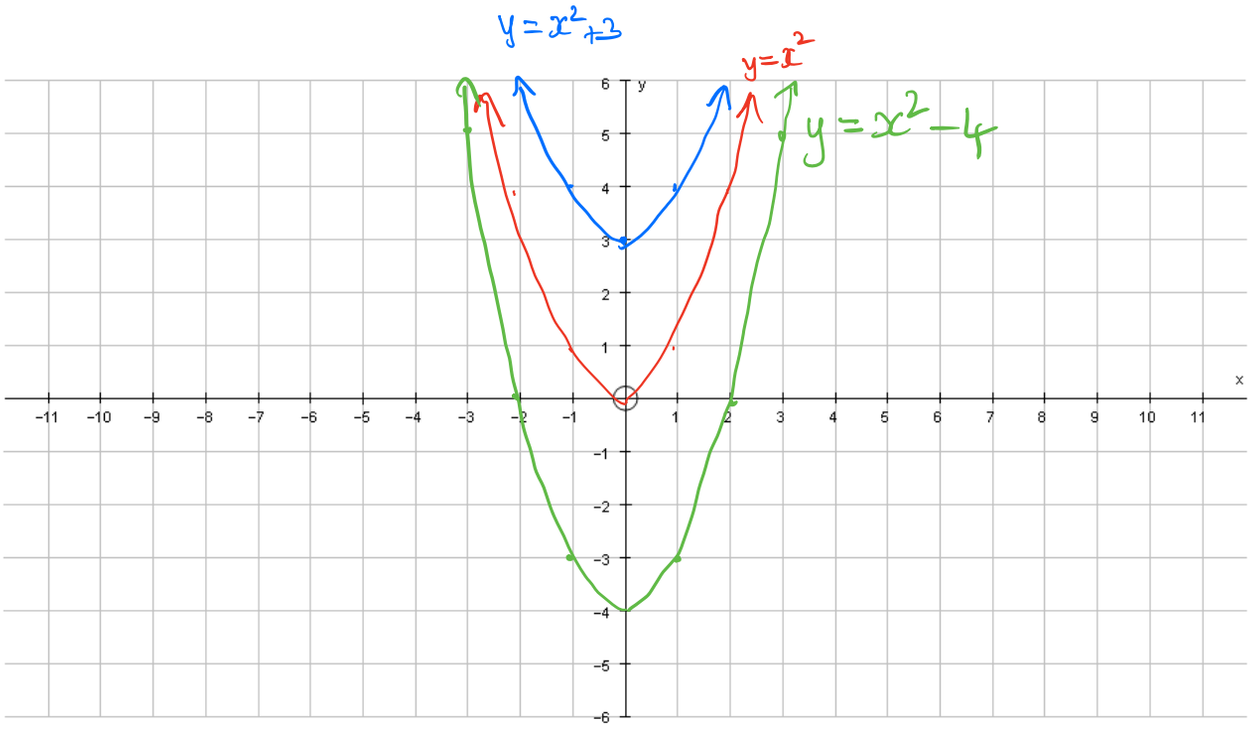
$= \frac{16 - 4}{4 - 2}$

$= 6$

B. **Sketch the following graphs on the same set of axes.**
Use different colours to represent each graph

i. $y = x^2$ ii $y = x^2 + 3$ iii. $y = x^2 - 4$

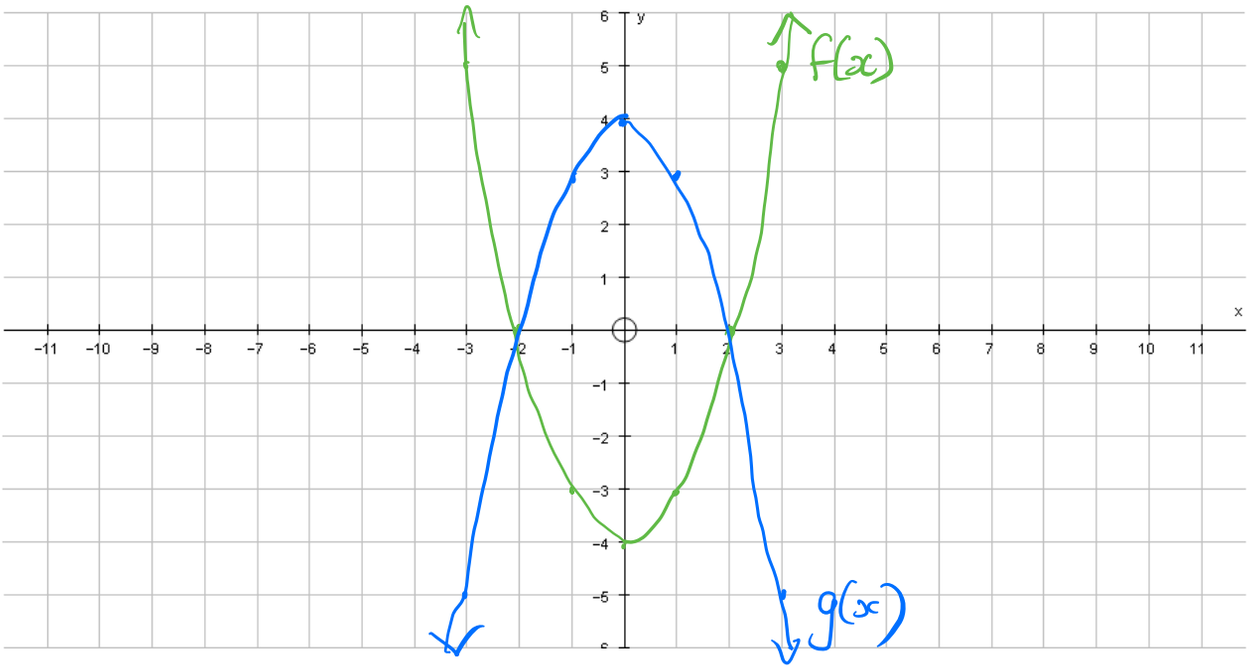
x	-3	-2	-1	0	1	2	3
$y = x^2$	9	4	1	0	1	4	9
$y = x^2 + 3$	12	7	4	3	4	7	12
$y = x^2 - 4$	5	0	-3	-4	-3	0	5



How does the value of q affect the graph? If $q > 0$ graph shifts up. If $q < 0$ graph shifts down

	$y = x^2$	$y = x^2 + 3$	$y = x^2 - 4$
Value of *a*	1	1	1
X - intercept	0	—	2 & -2
Y - intercept	0	3	-4
Domain	$x \in R$	$x \in R$	$x \in R$
Range	$y \in R; y \geq 0$	$y \in R; y \geq 3$	$y \in R; y \geq -4$
Co-ordinates of the turning point	(0;0)	(0;3)	(0;-4)
Axis of Symmetry	$x = 0$	$x = 0$	$x = 0$

1. a) On the given axes, sketch the graph of $f(x) = x^2 - 4$



b) i) On the same axes, sketch the graph $g(x)$ which is a reflection of $f(x)$ about the x-axis.

ii) Give the equation of $g(x)$ $= -x^2 + 4$

SUMMARY OF PARABOLA

The equation of any parabola has a standard form of:

$y = ax^2 + q$ or $f(x) = ax^2 + q$

a affects the rate of change (negative or positive) and whether the graph has a maximum or minimum value.

If $a > 0$ then the graph has a minimum value

If $a < 0$ then the graph has a maximum value

q affects the y-intercept, the point where the graph intersects with the y-axis and the y co-ordinate of the turning point.
The line $x = 0$ is the equation of the axis of symmetry.

SKETCHING A PARABOLA

Note restriction on domain of $f(x)$

EXAMPLE 1 : $f(x) = x^2 - 9;\ x \in (-\infty; 4)$ (Standard form)

1. Determine if graph has a max/min value. $a = 1 \therefore$ min value ∪

2. Find *y*-intercept (make *x*=0)

$y = 0^2 - 9$

$y = -9$

$(0 : -9)$

3. Find *x*-intercepts (make *y*=0)

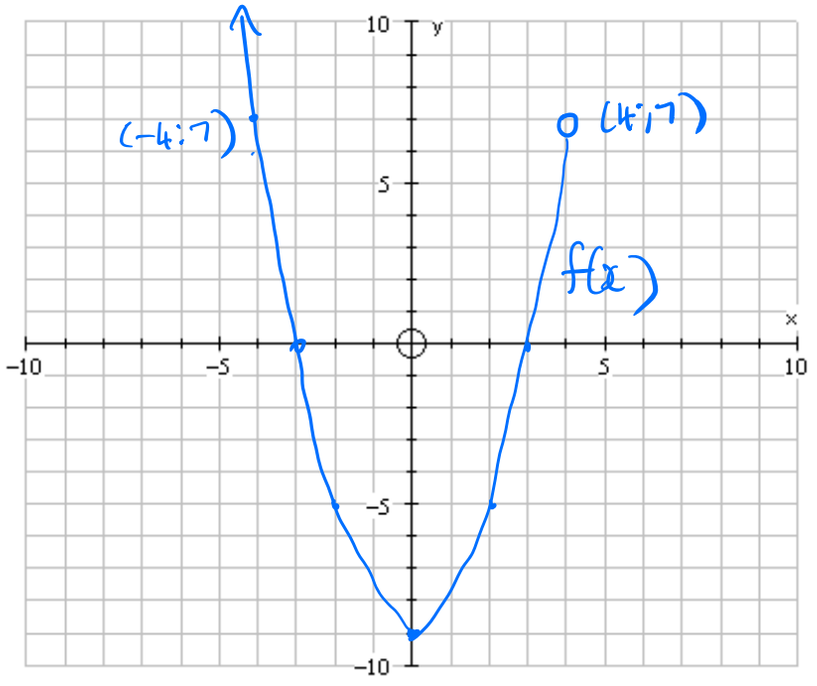
$0 = x^2 - 9$

$0 = (x+3)(x-3)$

$x = -3$ or $x = 3$

$(-3 : 0)$ or $(3 : 0)$

NB: Always write the coordinates of the *x*-intercepts (if there are any), the *y*-intercept and turning point.



Now do MAS page 149 Ex 10 (Nos 1 and 2)

Geogebra links for Parabola Graphs:

https://www.geogebra.org/m/BXWJB4y8 Quadratic Functions (Graph & Equation Anatomy)

https://www.geogebra.org/m/RaEvWF2n Basic Quadratic Function

Hyperbola

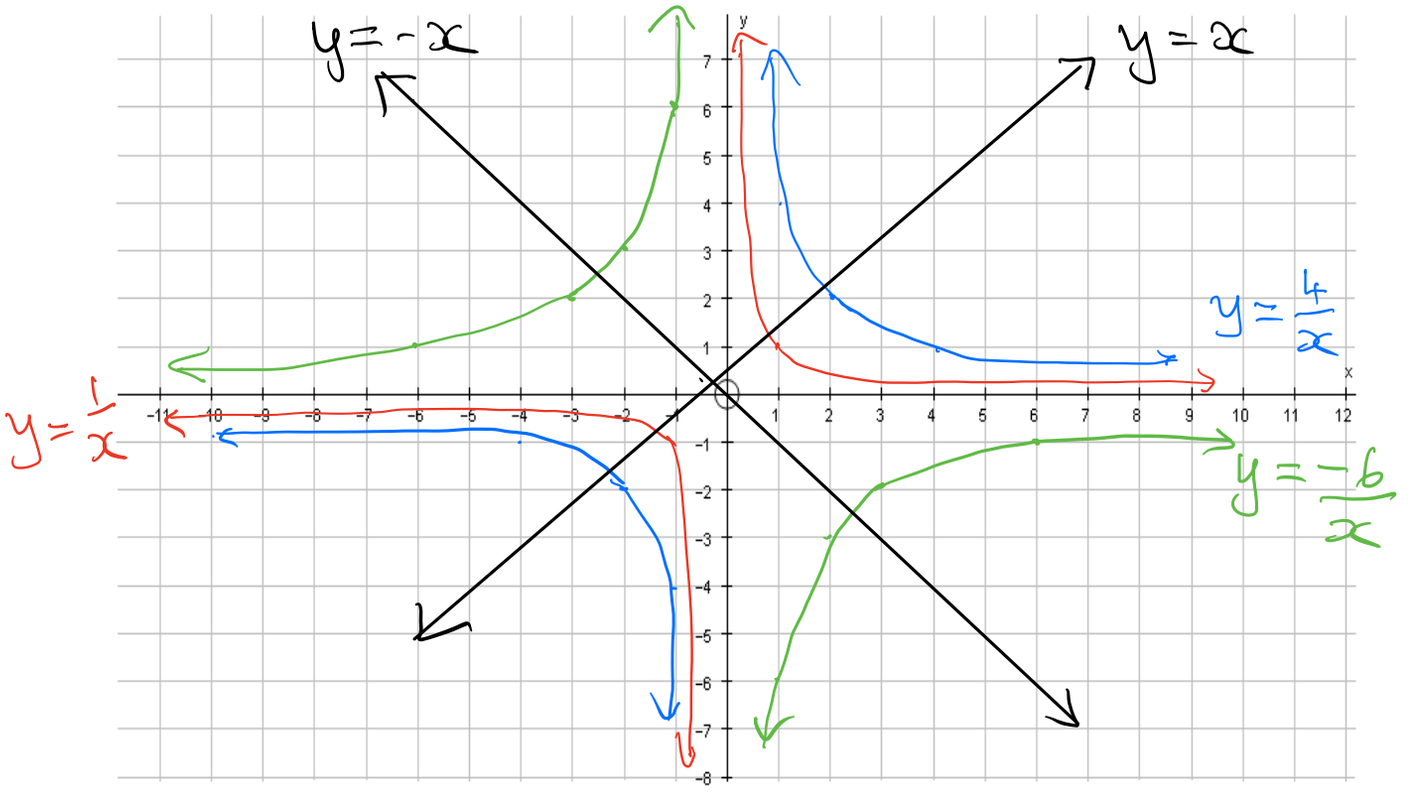
$y = \frac{a}{x}$ $\qquad\qquad$ $y = \frac{a}{x} + q$

The hyperbola is a **discontinuous** function. This means that the graph has two separate sections.

A. Sketch the following graphs on the same set of axes.
Use different colours to represent each graph

i. $y = \frac{1}{x}$ ii $y = \frac{4}{x}$ iii. $y = \frac{-6}{x}$

x	-3	-2	-1	0	1	2	3
$y = \frac{1}{x}$	$-\frac{1}{3}$	$-\frac{1}{2}$	-1	undef	1	$\frac{1}{2}$	$\frac{1}{3}$
$y = \frac{4}{x}$	$-\frac{4}{3}$	-2	-4	undef	4	2	$\frac{4}{3}$
$y = \frac{-6}{x}$	2	3	6	undef	-6	-3	-2



How does the value of ***a*** affect the graph?

When $a > 0$, graph lies in quads 1, 3

When $a < 0$, graph lies in quads 2, 4

Asymptote: This is a straight line that a curve approaches close to, but never reaches.
Notice that the above graphs approach the lines x = 0 and y = 0 (y- axis and x-axis)
Therefore we say that the **ASYMPTOTES** of the graph are **x = 0** and **y = 0**.
Indicate the asymptotes on the graph using a dotted line.

	$y=\frac{1}{x}$	$y=\frac{4}{x}$	$y=\frac{-6}{x}$
Value of *a*	1	4	-6
X - intercept	—	—	—
Y - intercept	—	—	—
Domain	$x\in R; x\neq 0$	$x\in R: x\neq 0$	$x\in R: x\neq 0$
Range	$y\in R: y\neq 0$	$y\in R: y\neq 0$	$y\in R: y\neq 0$
Asymptotes	$y=0, x=0$	$y=0, x=0$	$y=0, x=0$
Salient Point	$(1;1); (-1;-1)$	$(2;2)\ (-2;-2)$	$(-\sqrt{6};\sqrt{6})\ (\sqrt{6};-\sqrt{6})$

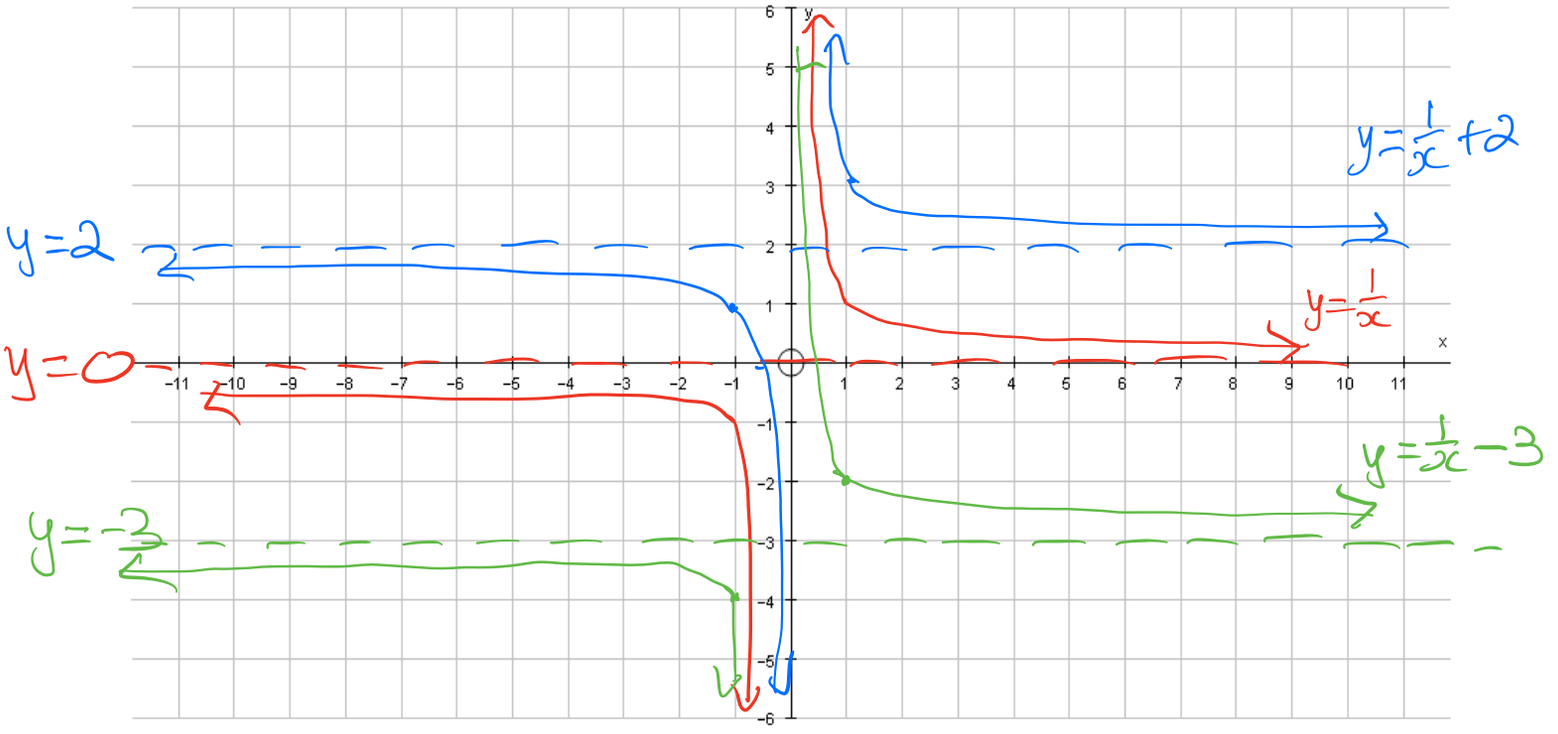
B. **Sketch the following graphs on the same set of axes.**
Use different colours to represent each graph

i. $y=\frac{1}{x}$ ii $y=\frac{1}{x}+2$ iii. $y=\frac{1}{x}-3$

x	-3	-2	-1	0	1	2	3
$y=\frac{1}{x}$	$-\frac{1}{3}$	$-\frac{1}{2}$	-1	undef	1	$\frac{1}{2}$	$\frac{1}{3}$
$y=\frac{1}{x}+2$	$1\frac{2}{3}$	$1\frac{1}{2}$	1	undef	3	$2\frac{1}{2}$	$2\frac{1}{3}$
$y=\frac{1}{x}-3$	$-3\frac{1}{3}$	$-3\frac{1}{2}$	-4	undef	-2	$-2\frac{1}{2}$	$-2\frac{2}{3}$

Indicate your asymptotes using a dotted line in the colour you have used for each graph

$x=0$



	$y=\frac{1}{x}$	$y=\frac{1}{x}+2$	$y=\frac{1}{x}-3$
Value of *a*	1	1	1
Value of q	0	2	-3
X - intercept	—	$-\frac{1}{2}$	$\frac{1}{3}$
Y - intercept	—	—	—
Domain	$x\in R; x\neq 0$	$x\in R: x\neq 0$	$x\in R: x\neq 0$
Range	$y\in R: y\neq 0$	$y\in R: y\neq 2$	$y\in R: y\neq -3$
Asymptotes	$x=0$; $y=0$	$x=0$; $y=2$	$x=0$; $y=-3$
Salient Point	(1;1) (-1;-1)	(1;3) (-1; 1)	(1;-2) (-1;-4)

How does the value of q affect the graph?

If $q>0$, shifts graph up

If $q<0$, shifts graph down

How have the asymptotes changed?

Vertical asymptote does not change

Horizontal asymptote shifts up/down depending on q value

SUMMARY OF HYPERBOLA

The equation of a hyperbola has a standard form of:

$$y = \frac{a}{x} + q \quad \text{or} \quad f(x) = \frac{a}{x} + q$$

q affects the vertical shift i.e. $\updownarrow$ movement and has an effect on the equation of the HORIZONTAL ASYMPTOTE (An asymptote is a line at which the graph is undefined)

The equation of the VERTICAL ASYMPTOTE is $x = 0$.

The vertical and horizontal asymptotes form the "quadrants" in which the hyperbola lies.

a affects the quadrants the hyperbola will lie in.

If $a > 0$ then the graph lies in quadrants 1 and 3

If $a < 0$ then the graph lies in quadrants 2 and 4

Vertical shift:

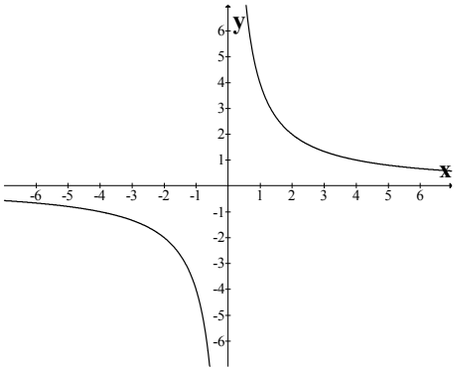
Adding a constant real number to the equation $y = \frac{a}{x}$ produces the equation $y = \frac{a}{x} + q$.

The y-values of $y = \frac{a}{x} + q$ and $y = \frac{a}{x}$ differ by a value of q.

The graph of $y = \frac{a}{x} + q$ will be shifted vertically upward from $y = \frac{a}{x}$ if $q > 0$ and shifted vertically downward if $q < 0$.

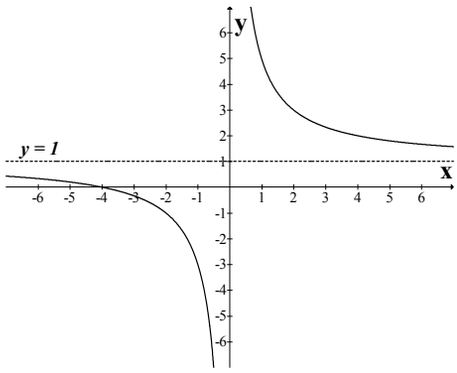
The position of the asymptote is also shifted by a value of q and is usually shown on a graph using a dotted line.

$y = \frac{4}{x}$



Equation of asymptote:
$y = 0$ and $x = 0$

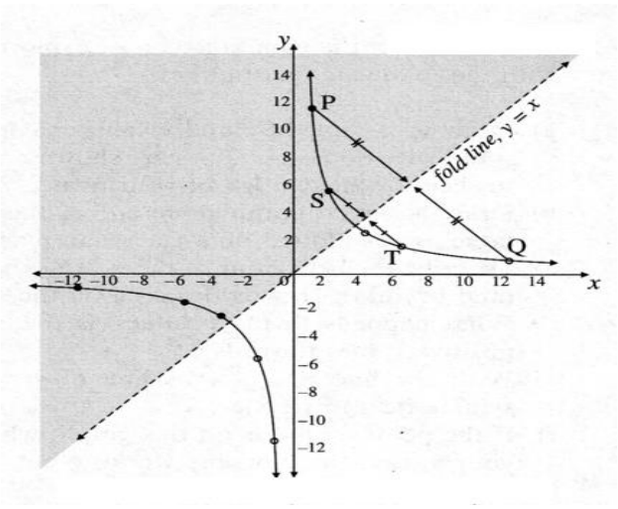
$y = \frac{4}{x} + 1$



Equation of asymptote:
$y = 1$ and $x = 0$

AXES OF SYMMETRY:

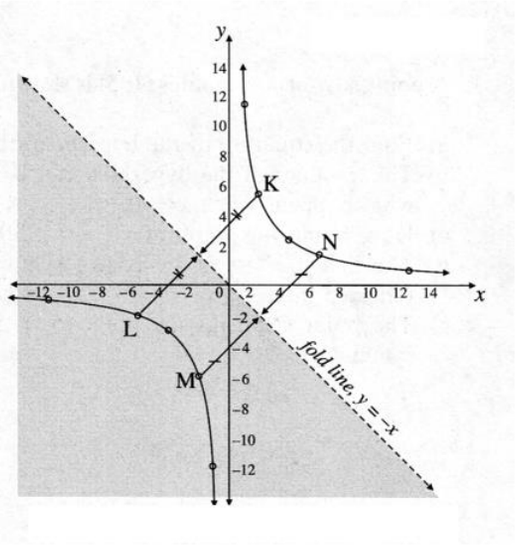
Example: $y = \frac{12}{x}$



The line $y = x$ is called the axis of symmetry.
Points P and Q are equal distances from the line $y = x$.

The coordinates of these points are interchanged.
If P (1;12) then the coordinates of Q are (12;1)

Points S and T are equal distances from the line $y = x$.
The coordinates are interchanged as follows:
S (2 ; 6) and T (6 ; 2)



The line $y = -x$ also divides the hyperbola into two identical mirror images.

Points K and L are examples of mirror images reflected about the line $y = -x$.
Their perpendicular distances to the line $y = -x$ are the same.

The coordinates of the mirror images are interchanged and the signs are opposite.
If K (2 ; 6) then L $(-6;-2)$

Geogebra links for Hyperbola Graphs:

https://www.geogebra.org/m/FHFGwJZN Basic rectangular hyperbola and translations.

https://www.geogebra.org/m/eTFVMsgH
Use the sliders a and k to explore the effect of the parameters in the equation of a **hyperbola**.
Let slider h = 0. In grade 11 we will explore the effects of this slider.

SKETCHING A HYPERBOLA

EXAMPLE : $f(x)=\frac{4}{x}+3$ and $g(x)=\frac{-8}{x}$

1. Determine which quadrants the graph will lie in. $f(x)$ lies in Q 1, 3 $(a=4)$
   $g(x)$ lies in Q 2, 4 $(a=-8)$
2. Draw the positions of the asymptotes.
3. Find x-intercepts (make $y = 0$).

Graph f: $0=\frac{4}{x}+3$

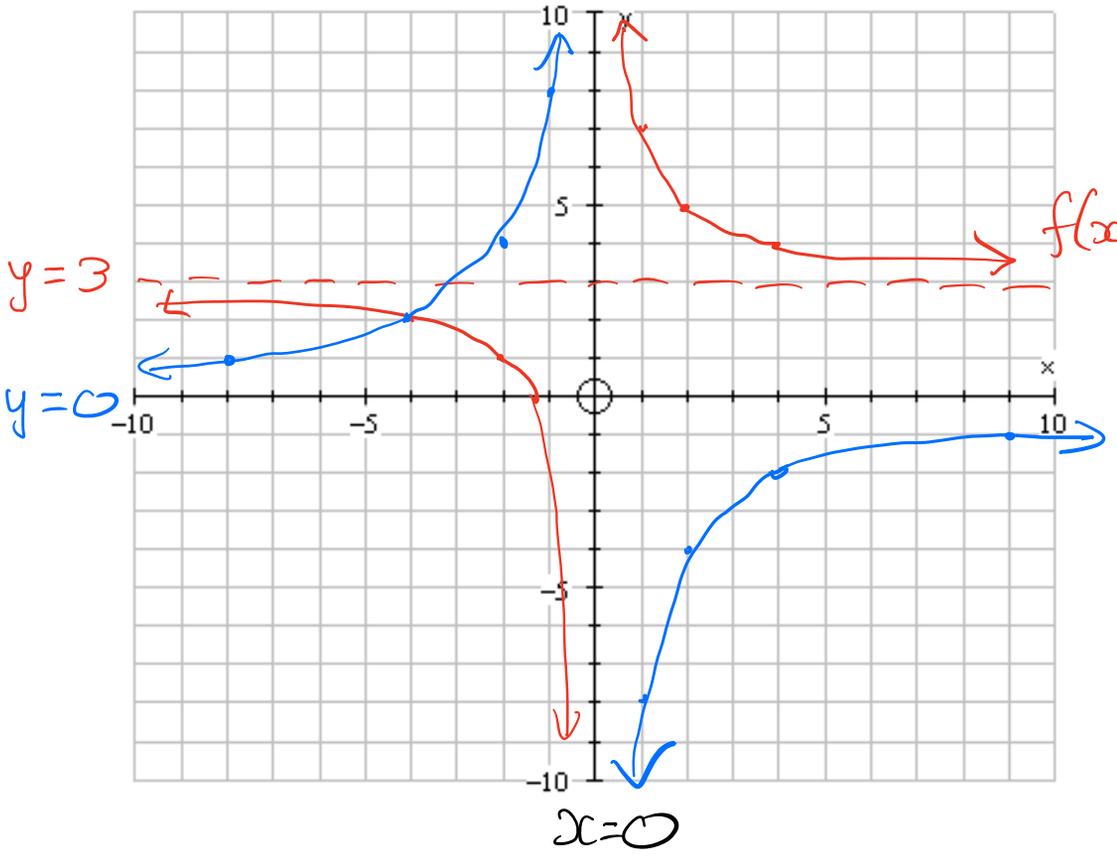
$-3=\frac{4}{x}$

$x=4/-3$

Graph g: $0=\frac{-8}{x}$

no solution

$\therefore$ no x-intercept



EQUATIONS OF AXES OF SYMMETRY:

$f(x)=\frac{4}{x}+3$ $y=x+3$ $y=-x+3$

$g(x)=\frac{-8}{x}$ $y=x$ $y=-x$

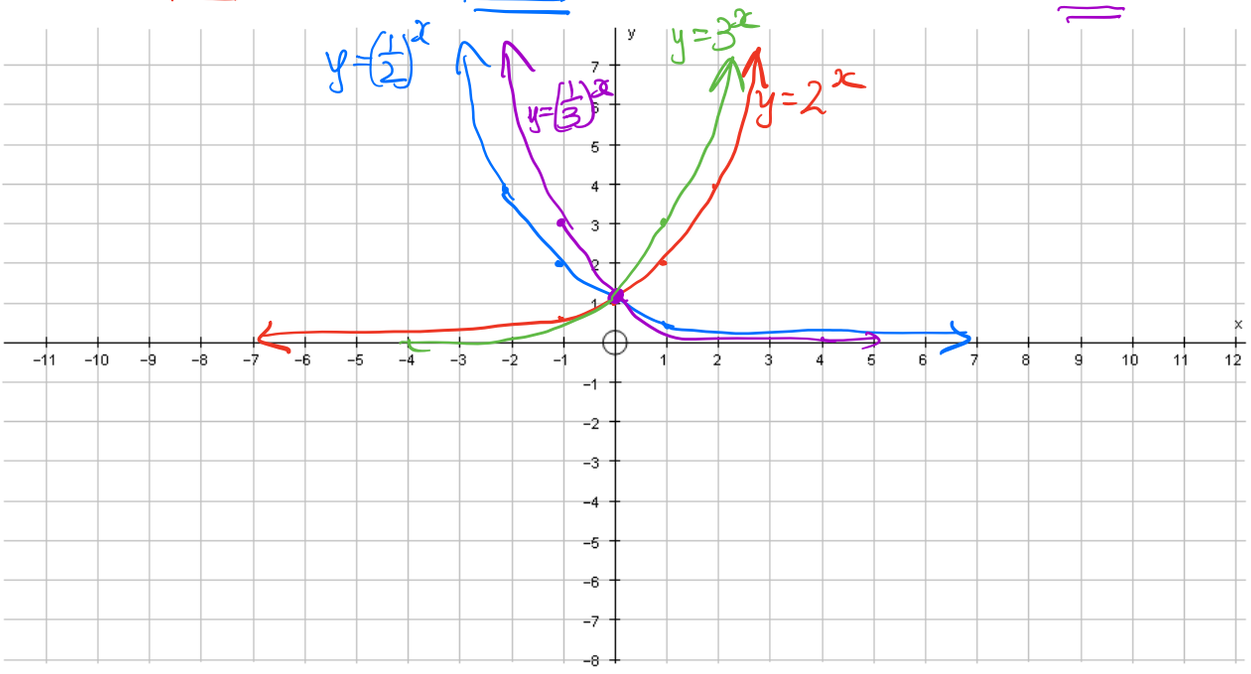
Now do: MAS page 159 Ex 13 (No 1.b,g).

Exponential function

$y = a.b^x$ $\qquad\qquad$ $y = a.b^x + q$

A. **Sketch the following graphs on the same set of axes.**
Use different colours to represent each graph

i. $y = 2^x$ ii $y = \left(\frac{1}{2}\right)^x$ iii. $y = 3^x$ iv. $y = \left(\frac{1}{3}\right)^x$



	$y = 2^x$	$y = \left(\frac{1}{2}\right)^x$	$y = 3^x$	$y = \left(\frac{1}{3}\right)^x$
Value of a	1	1	1	1
Value of b	2	$\frac{1}{2}$	3	$\frac{1}{3}$
Y - intercept	1	1	1	1
Domain	$x \in R$	$x \in R$	$x \in R$	$x \in R$
Range	$y \in R$; $y > 0$	$y \in R$; $y > 0$	$y \in R$; $y > 0$	$y \in R$; $y > 0$
Asymptotes	$y = 0$	$y = 0$	$y = 0$	$y = 0$
Growth or Decay function	G	D	G	D

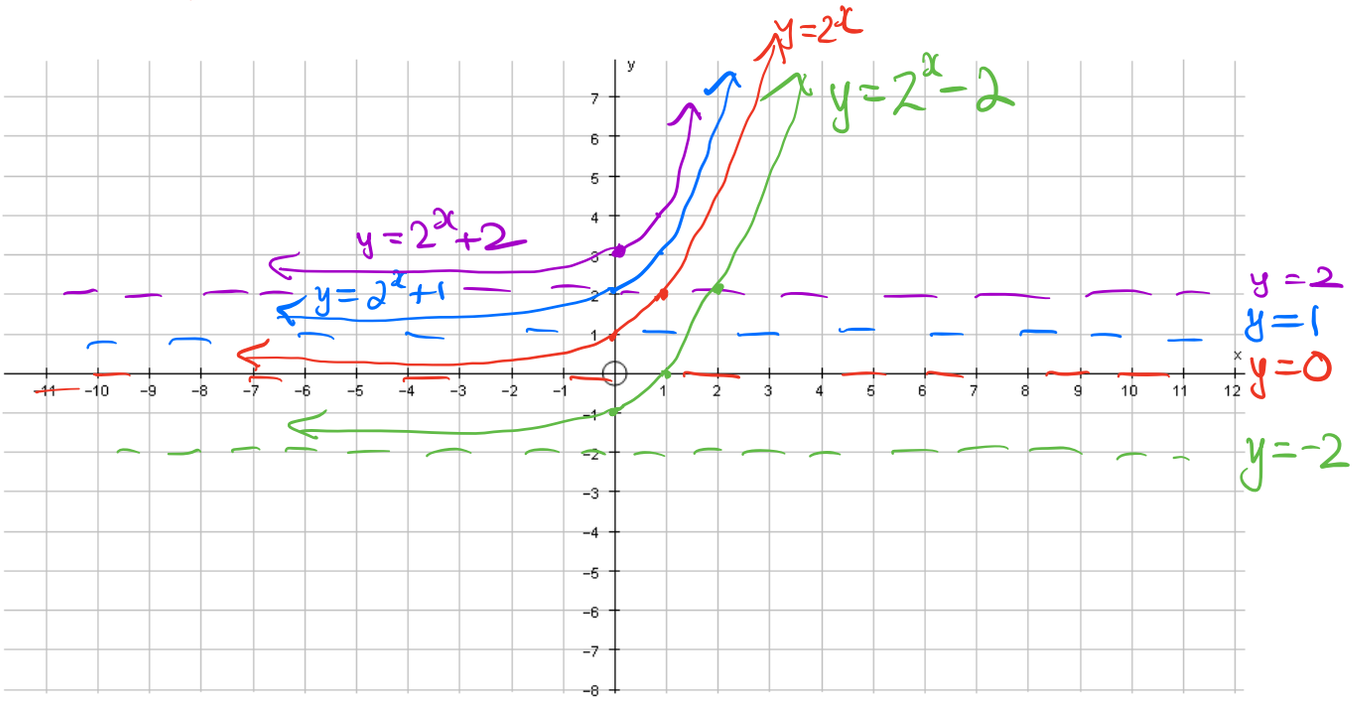
What effect does *b* have on the graph? If $a > 0$ *

If $b > 1$ increasing function

If $0 < b < 1$ decreasing function

B. **Sketch the following graphs on the same set of axes.**
Use different colours to represent each graph

i. $y = 2^x$ ii $y = 2^x + 1$ iii. $y = 2^x + 2$ iv. $y = 2^x - 2$



	$y = 2^x$	$y = 2^x + 1$	$y = 2^x + 2$	$y = 2^x - 2$
Value of a	1	1	1	1
Value of b	2	2	2	2
Value of q	0	1	2	-2
Y - intercept	1	2	3	-1
Domain	$x \in R$	$x \in R$	$x \in R$	$x \in R$
Range	$y \in R_{y>0}$	$y \in R_{y>1}$	$y \in R_{y>2}$	$y \in R_{y>-2}$
Asymptotes	$y = 0$	$y = 1$	$y = 2$	$y = -2$
Growth or Decay function	G	G	G	G

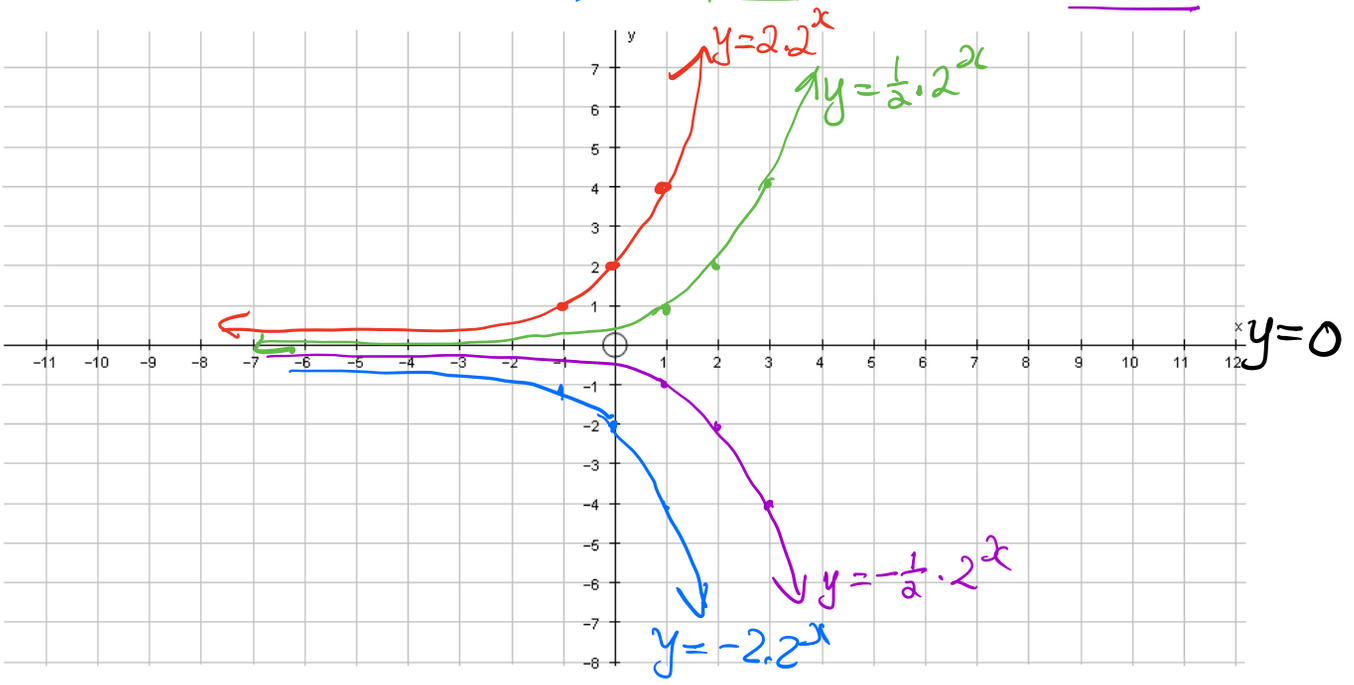
What effect does *q* have on the graph?

If $q > 0$, graph shifts up

If $q < 0$, graph shifts down

C. **Sketch the following graphs on the same set of axes.**
Use different colours to represent each graph

i. $y = 2.2^x$ ii $y = -2.2^x$ iii. $y = \frac{1}{2}.2^x$ iv. $y = -\frac{1}{2}.2^x$



	$y = 2.2^x$	$y = -2.2^x$	$y = \frac{1}{2}.2^x$	$y = -\frac{1}{2}.2^x$
Value of *a*	2	-2	½	-½
Value of b	2	2	2	2
Y - intercept	2	-2	½	-½
Domain	$x \in R$	$x \in R$	$x \in R$	$x \in R$
Range	$y \in R; y > 0$	$y \in R; y < 0$	$y \in R; y > 0$	$y \in R; y < 0$
Asymptotes	$y = 0$	$y = 0$	$y = 0$	$y = 0$
Growth or Decay function	G	D	G	D

What effect does *a* have on the graph?

It stretches / compresses the graph
It affects the y-intercept / increase or decrease of the function / reflection over asymptote

Are the graphs $y = 2.2^x$ and $y = -2.2^x$, reflections of each other about the x axis? Yes

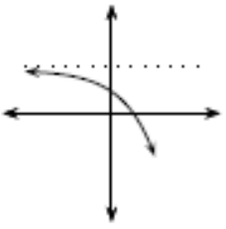
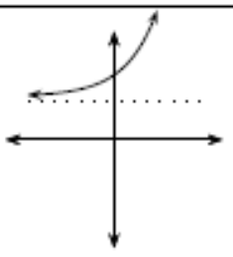
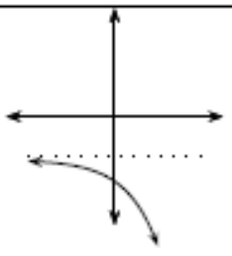
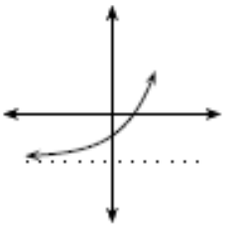
How could you tell this without looking at the graphs? value of 'a' opposite

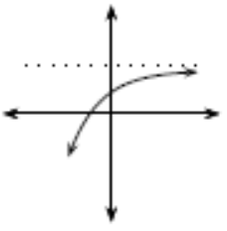
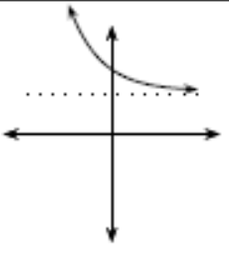
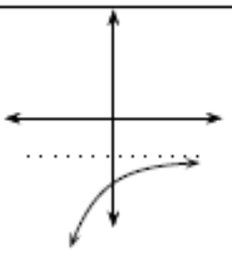
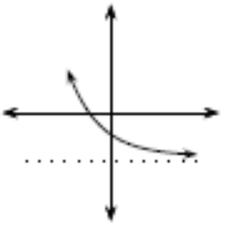
SUMMARY OF EXPONENTIAL

The equation of an exponential graph has a standard form of:

$$y = ab^x + q \quad \text{or} \quad f(x) = ab^x + q$$

where $b > 0$ and $b \neq 1$

$b > 1$	$a < 0$	$a > 0$
$q > 0$		
$q < 0$		

$0 < b < 1$	$a < 0$	$a > 0$
$q > 0$		
$q < 0$		

NOTE: THERE IS NO AXIS OF SYMMETRY FOR EXPONTENTIAL FUNCTIONS

SKETCHING AN EXPONENTIAL GRAPH

EXAMPLE 1. Draw the graphs of: $y = 3.2^x$ and $y = -3^x + 2$

$a > 0 : b > 0 : q = 0$ $\quad a < 0 : b > 0 : q > 0$

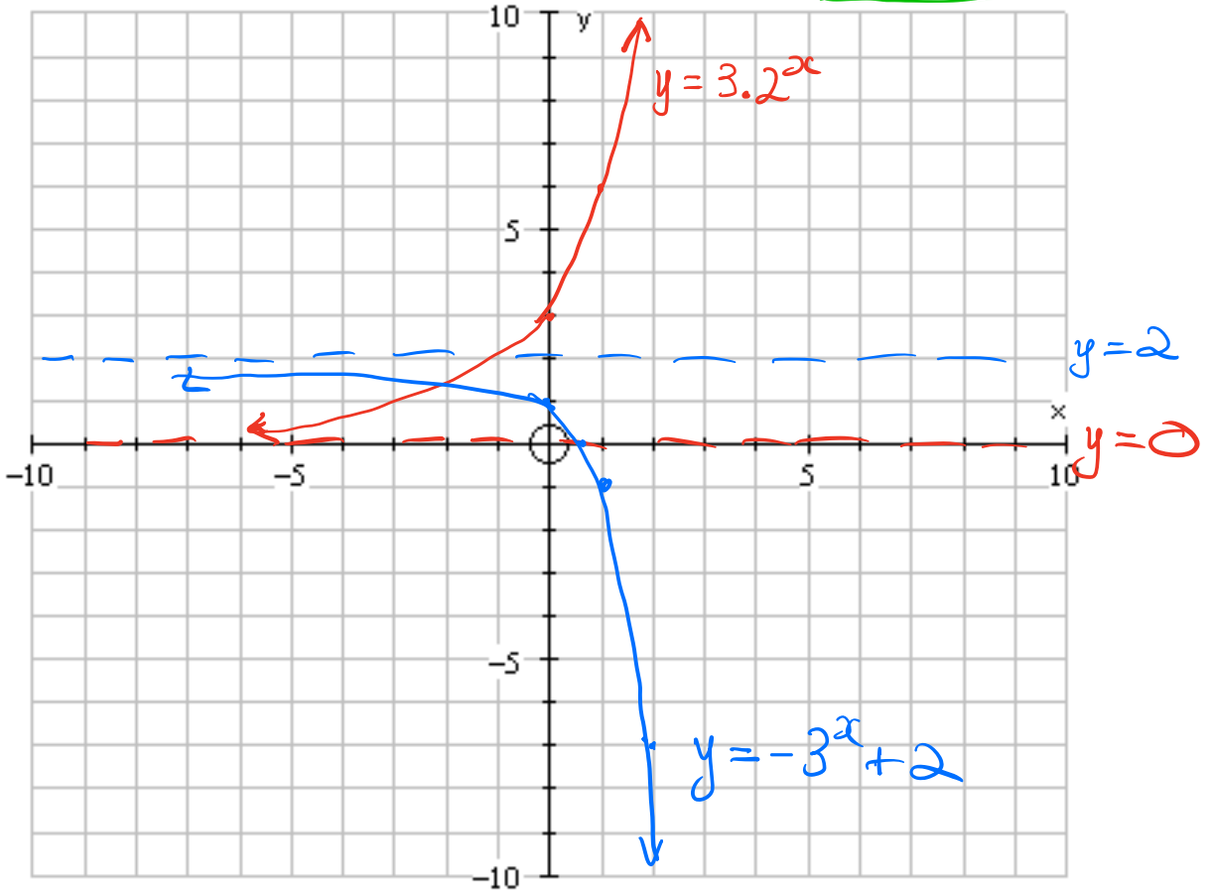
1. Decide whether the function is increasing or decreasing.
2. Plot the asymptote.
3. Find the *y*-intercept (make $x = 0$).

   $y = 3 \times 2^0 = 3$ and $y = -3^0 + 2 = 1$
4. Find the x-intercept (make $y = 0$).

   $0 = 3.2^x \therefore$ no x-int

   $0 = -3^x + 2$

   $3^x = 2^1 \quad x = 0 \; ; 63$



Now do: MAS page 171 Ex 17 (No 1b,e)

Geogebra links for Exponential Graphs:

https://www.geogebra.org/m/YK6pcUsB

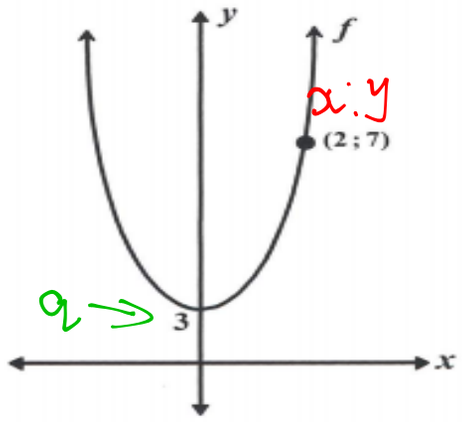
https://www.geogebra.org/m/vvXYtCfr

https://www.geogebra.org/m/mwJkxwS6

EQUATION OF GRAPHS

1. Parabola

a) Determine the equation of the graph in the form $f(x) = ax^2 + q$



$y = ax^2 + 3$

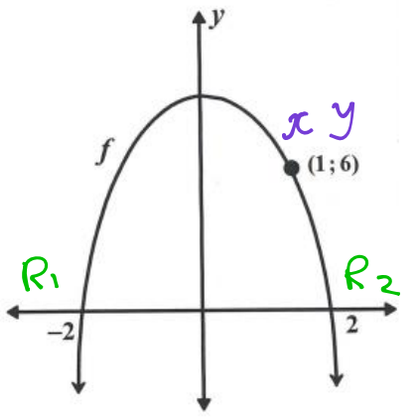
$7 = a(2)^2 + 3$

$4 = 4a$

$1 = a$

$\therefore f(x) = x^2 + 3$

b) Determine the equation of the graph in the form $f(x) = ax^2 + q$



$y = a(x - R_1)(x - R_2)$

$y = a(x - -2)(x - 2)$

$6 = a(1 + 2)(1 - 2)$

$6 = -3a$

$-2 = a$

$y = -2(x+2)(x-2)$

$y = -2(x^2 - 4)$

$y = -2x^2 + 8$

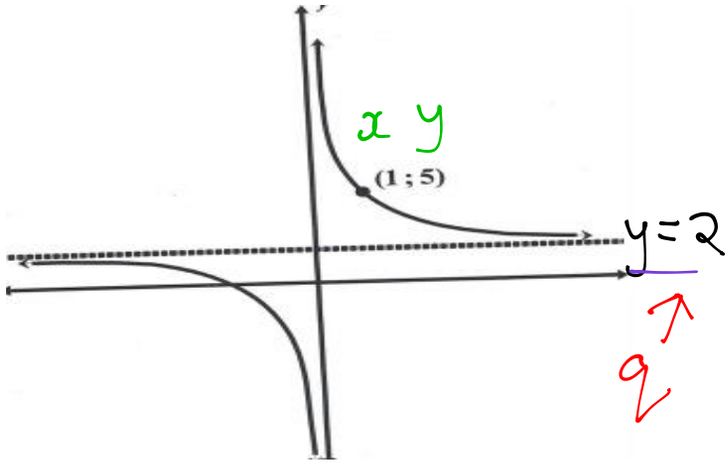
$f(x) = -2x^2 + 8$

Complete Exercise 11 p150 (MAS)

2. Hyberbola

Determine the equation of the graph in the form

$$f(x) = \frac{a}{x} + q$$



$y = \frac{a}{x} + 2$

$5 = \frac{a}{1} + 2$

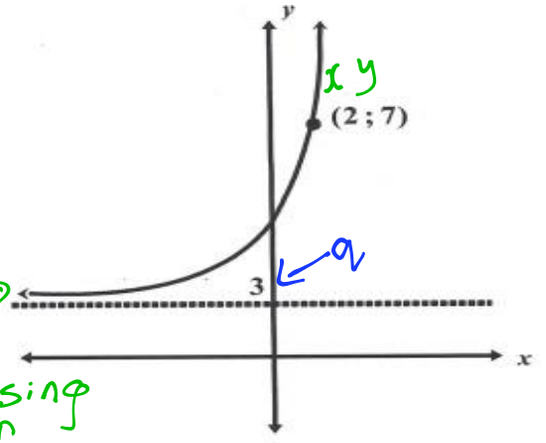
$3 = a$

$f(x) = \frac{3}{x} + 2$

COMPLETE Exercise 14 P161 (MAS)

3. Exponential

Determine the equation of the graph in the form $f(x) = a^x + q$



$y = a^x + 3$

$7 = a^2 + 3$

$4 = a^2$

$0 = a^2 - 4$

$0 = (a+2)(a-2)$

$a = -2$ or $a = 2$

But $a > 1$ or $0 < a < 1$

$f(x) = 2^x + 3$

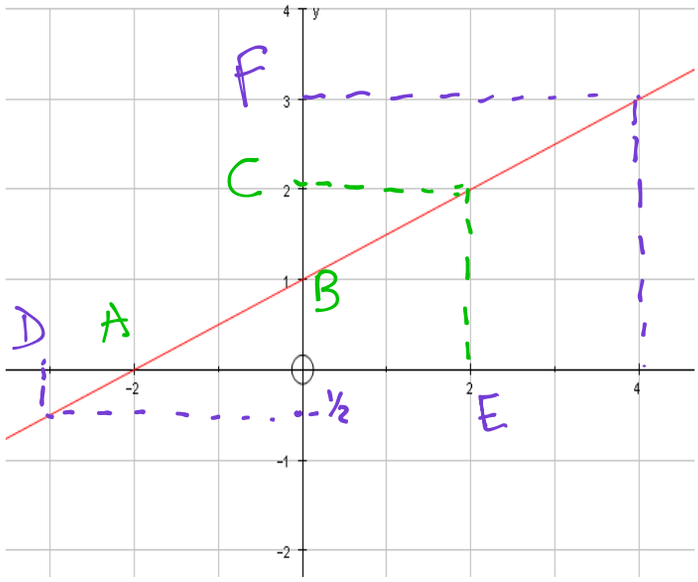
COMPLETE Exercise 18 p173 MAS

FUNCTION APPLICATIONS

1. **"READ OFF THE GRAPH"**

Example:
The graph represents the equation of $y = \frac{1}{2}x + 1$.

Use the graph to answer the following questions. Use the letters A; B; C etc to show where readings are made



a) Give the x-intercept ___-2___ (Read off at ___A___)

b) Give the y-intercept ___1___ (Read off at ___B___)

c) If $x = 2$, give the corresponding value of y ___2___ (Read at ___C___)

d) If $y = -\frac{1}{2}$, give the corresponding x value ___-3___ (Read at ___D___)

e) Find the value of x, if $\frac{1}{2}x + 1 = 2$ ___$\therefore x = 2$___ (Read at ___E___)

f) If $x = 4$, give the corresponding value of y. ___3___ (Read at ___F___)

2. FUNCTION NOTATION

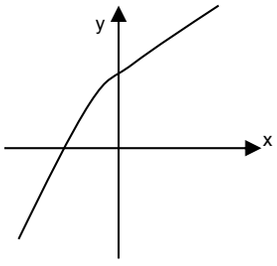
A **function** is a relation where for each x there is only one y.

Example:

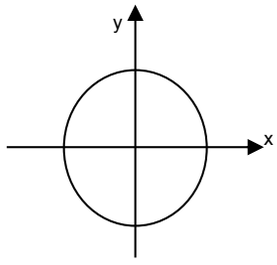
$y = x^2 + 1$ is a function

$y = \pm\sqrt{x}$ is not a function

A function



Not a function



Vertical line test: Move a line vertically through the graph. If it ever cuts the graph twice, then it is **not** a function.

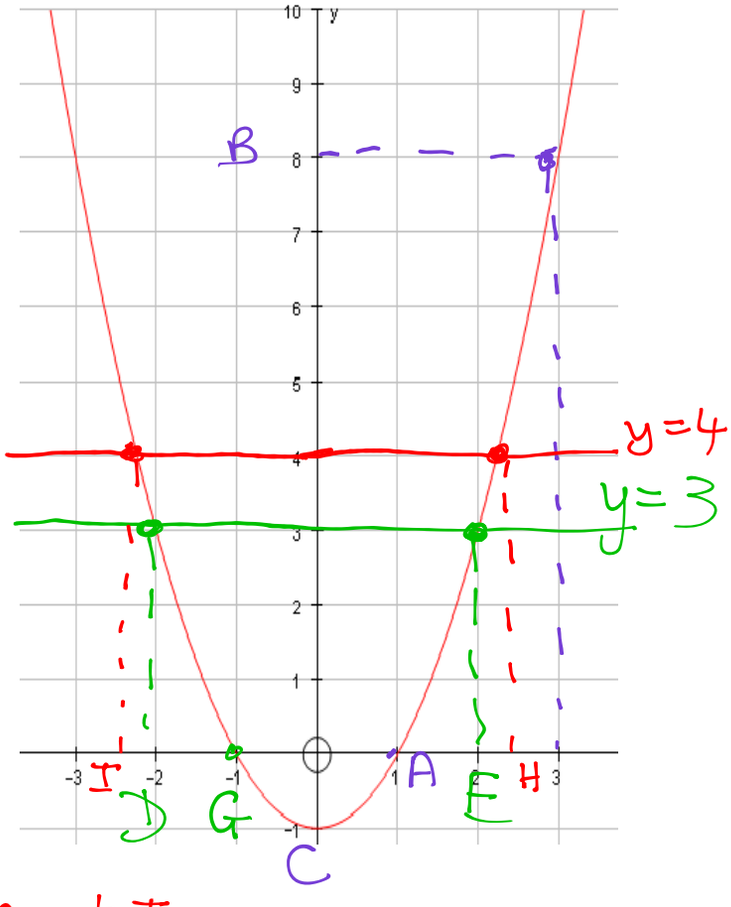
If y is a function of x and $\boldsymbol{y = -2x + 4}$.
Then you can write the function as $\boldsymbol{f(x) = -2x + 4}$ **OR** $\boldsymbol{f: x \to -2x + 4}$

If $f(x) = -2x + 4$, *then* $f(-2) = -2(-2) + 4 = 8$

$f(x) > 0$ *x- values above* x *–axis* $(y = 0)$
$f(x) < 0$ *x - values below* x*-axis* $(y = 0)$
$f(x) > 1$ *x- values above the horizontal line of* $y = 1$.

Example:

Graph of $f: x \to x^2 - 1$ is given.



Use the graph to find the following:

a) $f(1)$ = 0 (Read at A)

$f(3)$ = 8 (Read at B)

$f(0)$ = −1 (Read at C)

b) x so that $f(x) = 3$

$x = -2$ or $x = 2$

at D at E

c) the zero *of* f: (x-intercepts)

$x = -1$ at G

$x = 1$ at A

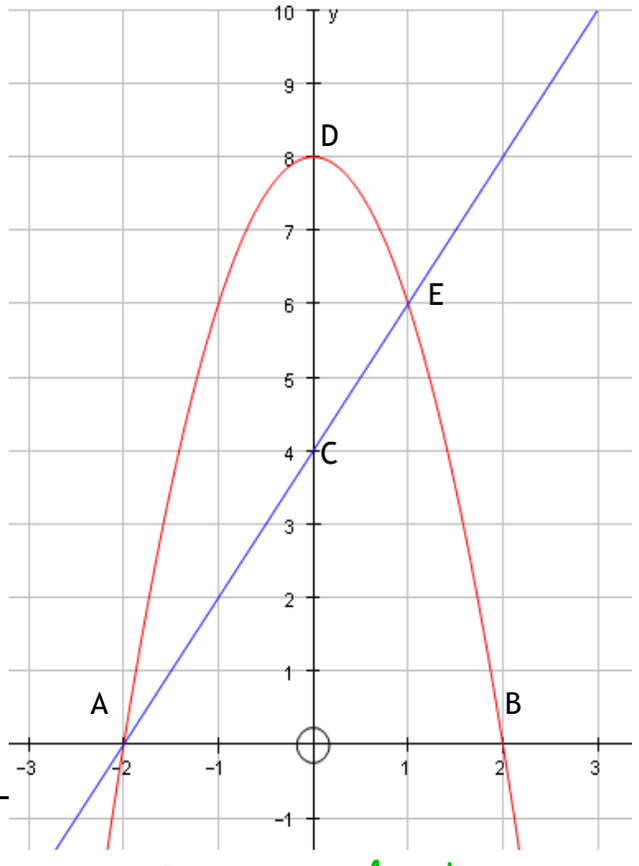
d) a such that $f(a) = 4$ $a = -2{,}2$ at I

or $a = 2{,}2$ at H

4. LENGTH

NB: Length is always positive

Example: **$y = -2x^2 + 8$ is given**



Do not rely on the grid to determine the answers

1. **Find the length of OA.**
   A is the x-intercept.
   Therefore, we need to find the x-int. (y = 0)

$0 = -2x^2 + 8$

$2x^2 = 8$

$x^2 = 4 \therefore x = \pm 2$

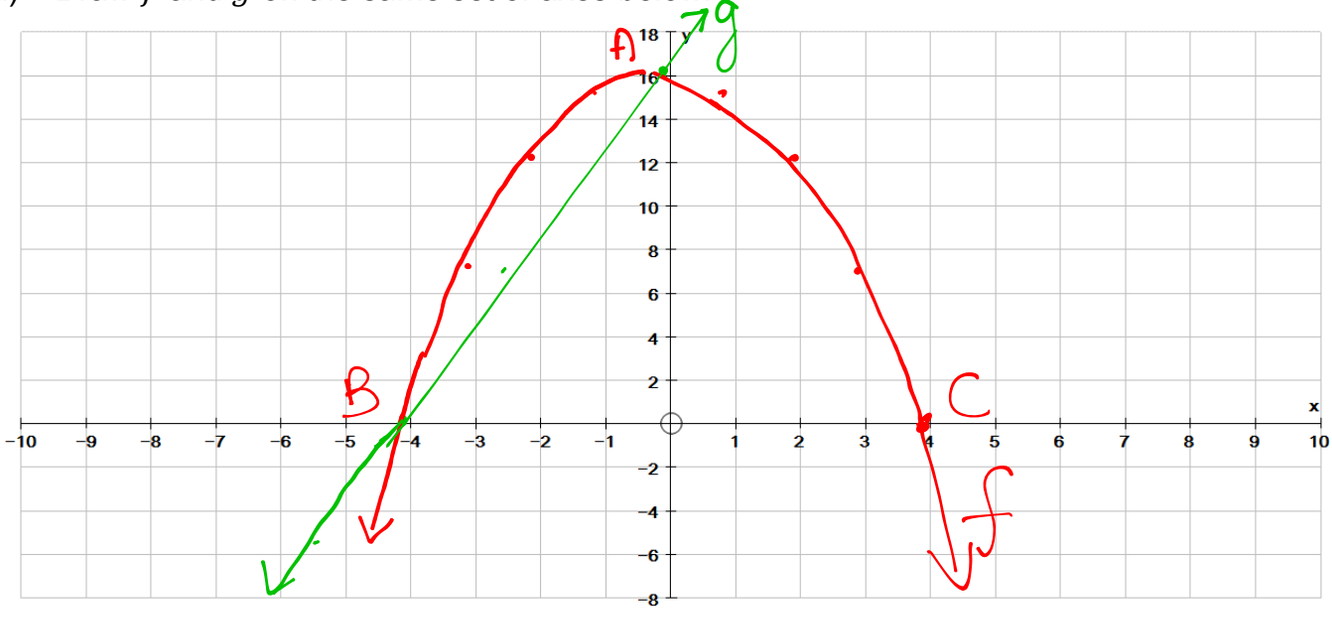
Length OA = 2 units

2. **Length of AB.** 4 (2 + 2)
3. **Length of OC.** 4 (y-int C)
4. **Length of CD.** 4 (8 − 4) Difference between the two y-intercepts (at D and C)

CONSOLIDATION EXERCISES

1. Given the functions $f(x) = -x^2 + 16$ and $g(x) = 4x + 16$

   (a) Draw f and g on the same set of axes below.



(b) Use your graphs to solve for x if:

(1) $f(x) = g(x)$ at -4 (B) and 0 (A)

(2) $f(x) \geq 0$ $-4 \leq x \leq 4$ (from B to C)

(c) Give the equation of the reflection of f in the x-axis.

$-y = -x^2 + 16 \quad \therefore \quad y = x^2 + 16$

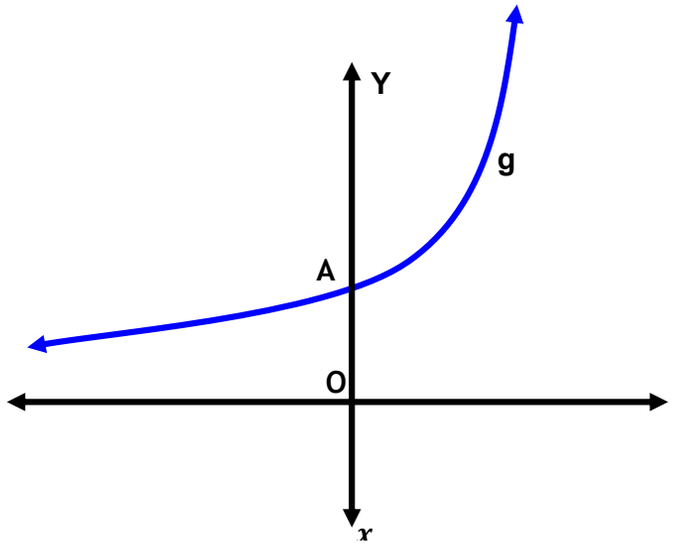
(d) Write down the co-ordinates of the turning point of f.

(0; 16)

(e) Does f have a maximum or a minimum value? maximum

What is this value? 16

2. The graph of $g(x) = 2^x$ is shown cutting the y-axis at A.



Answer the following questions by using the graph.

(a) Write down the co-ordinates of point A. (0; 1) y-intercept of g(x)

(b) Give the equation of the asymptote of the graph of $y = 2^x - 3$ $y = -3$

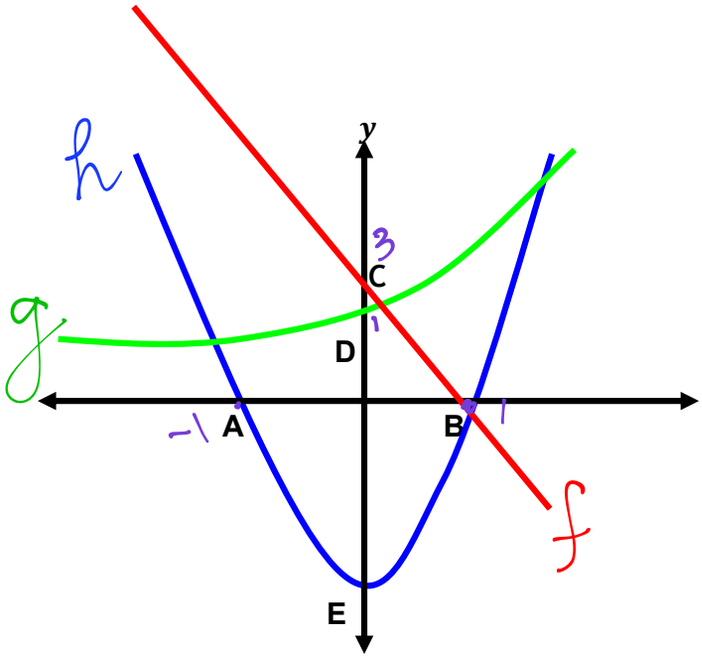
(c) For which value of x will $g(x) = \frac{1}{8}$?

$2^x = \frac{1}{8} \quad \therefore \quad 2^x = 2^{-3} \quad \therefore \quad x = -3$

(d) Give the domain, and the range of $y = 2^x$.

Domain: $\{x \in R\}$ Range $\{y \in R : y > 0\}$

3.



In the diagram, which is not drawn to scale, are the graphs of:

$f(x) = 3 - 3x$, $g(x) = 3^x$ and $h(x) = 3x^2 - 3$

Graph *f* cuts the y-axis at C and the x-axis at B, graph *g* cuts the y-axis at and graph *h* cuts the x-axis at A and B and the y-axis at E.

(a) Find the length of CD. $3 - 1 = 2$

(b) Find the length of DE. $1 + 3 = 4$

(c) Find the length of AB. $3x^2 - 3 = 0 \Rightarrow 3(x^2 - 1) = 0 \Rightarrow 3(x+1)(x-1) = 0$
$\therefore AB = 1 + 1 = 2$ (A) $x = -1$ or $x = 1$ (B)

(d) Find the equation of $k(x)$ which is the reflection of $h(x)$ in the x-axis.
$-y = 3x^2 - 3$
$\therefore k(x) = -3x^2 + 3$

(e) Find the equation of $q(x)$ which is the reflection of $f(x)$ in the line $y = x$ swop letters
$x = 3 - 3y$
$3y = 3 - x$ $\therefore q(x) = 1 - \frac{1}{3}x$

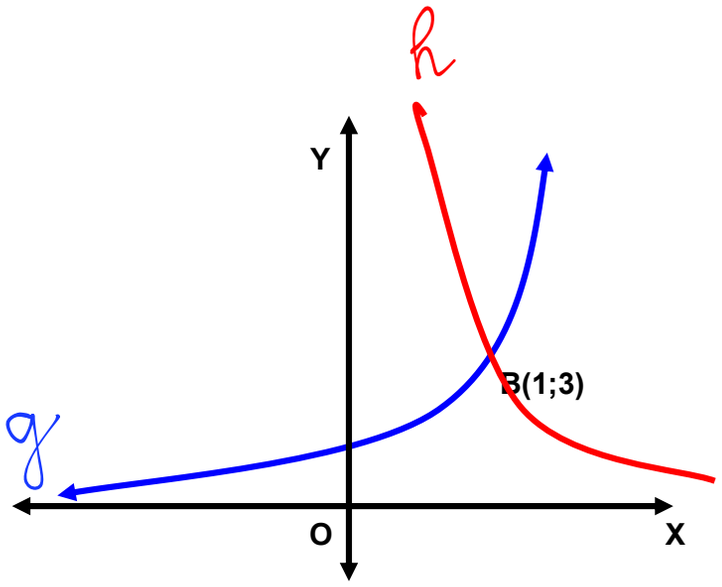
(f) Without changing the shape or symmetry about the y-axis of any of the graphs, they need to be moved so that they all have a y-intercept at the origin. Give the new equation for each of the original graphs.

f ' (x) = $-3x$

g' (x) = $3^x - 1$

h' (x) = $3x^2$

4.



The graph of $g(x) = 3^x$ and $h(x) = \frac{k}{x}$, $x > 0$, are drawn above and intersect at B(1;3).

(a) What is the domain of g? $x \in R$

(b) What is the range of h? $y \in R ; y \neq 0$

(c) What is the value of k? $3 = \frac{k}{1} \therefore k = 3$

(d) What is the new equation of g if reflected in the x-axis? $-y = 3^x \therefore g'(x) = -3^x$

(e) What is the new equation of g if reflected in the y-axis? $g'(x) = 3^{-x} = (3^{-1})^x = \left(\frac{1}{3}\right)^x$

(f) What is the new equation of h if reflected in $y = x$? $x = \frac{3}{y}, y > 0 \therefore y = \frac{3}{x}, y > 0$

(g) What is the asymptote of the graph $y = 3^x + 4$? $y = 4$

Mind Action Series

- **P175 – Mixed Revision Exercises**
- **P 179 Some Challenges**